Hints: Linear algebra – 1. Determinants and invertibility

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $B = A - J$, where J is the all-ones matrix, which has rank 1. Show that $\operatorname{rank} B^3 < 10$ (or $\operatorname{rank} B < 10$).

Hint 2. Expand $B^3 = (A - J)^3$. The term A^3 vanishes, and each of the remaining 7 terms contains a factor J, so has rank at most 1. Hence $\operatorname{rank} B^3 \leqslant 7 < 10$, so $\det B^3 = 0$.

Problem 2

Hint 1. Multiply the relation by $A+B$: from $E = (A+B)(A^{-1}+B^{-1})$ get $AB^{-1}+BA^{-1} = -E$.

Hint 2. Multiplying $AB^{-1} + BA^{-1} = -E$ by B on the right (or by A) gives $A + B = -BA^{-1}B$ and similarly $A + B = -AB^{-1}A$. Equate and take determinants: $(\det A)^3 = (\det B)^3$, which over $\mathbb{R}$ gives $\det A = \det B$.

Hint 3. Over $\mathbb{C}$ the cube roots of unity break this. Try scalar matrices $A = \omega E$, $B = \bar{\omega} E$ with $\omega = e^{i\pi/3}$ (check $\omega^{-1} + \bar{\omega}^{-1} = (\omega + \bar{\omega})^{-1}$) and $n = 2$: the answer is no.

Problem 3

Hint 1. The eigenvalues of M are the roots of $x^2 - ax + b$, and they are real since M is symmetric; so $a^2 \geqslant 4b$ is necessary for existence.

Hint 2. If the eigenvalues are distinct, $M = Q^T \operatorname{diag}(\lambda_1, \lambda_2) Q$ and rotating Q gives infinitely many such M. If $\lambda_1 = \lambda_2 = \lambda$, then $M = \lambda E$ is forced. Answer: exactly the pairs with $a^2 = 4b$.

Problem 4

Hint 1. 2) $\Rightarrow$ 1) is immediate (take $B_i = A_i$). For the converse, use that A_i is symmetric: $\langle (\sum A_i^2)x, x \rangle = \sum \|A_i x\|^2$.

Hint 2. Take $x \neq 0$ in the kernel of $\sum A_i^2$; then $A_i x = 0$ for every i. So x is in the kernel of $\sum B_i^T A_i = \left(\sum A_i B_i\right)^T$ for any B_i.

Problem 5

Hint 1. $S = \sum A_i^T A_i$ is symmetric and $\langle Sx, x \rangle = \sum \|A_i x\|^2 \geqslant 0$. What does this say about its eigenvalues?

Problem 6

Hint 1. A is singular, so $\operatorname{rank} \operatorname{adj} A \leqslant 1$.

Hint 2. For a matrix C of rank $\leqslant 1$, write $C = uv^T$ and use the rank-one update formula: $\det(E + C) = 1 + v^T u = 1 + \operatorname{tr} C$. Apply it to $C = \operatorname{adj} A$.

Problem 7

Hint 1. One direction is a computation: if $A = B^{-1}\bar{B}$ then $\bar{A} = \bar{B}^{-1}B = A^{-1}$. For the converse you need B invertible with $BA = \bar{B}$.

Hint 2. Try $B = zE + \bar{z}\,\bar{A}$ with $z \in \mathbb{C} \setminus \{0\}$. Using $\bar{A}A = E$, check that $BA = \bar{B}$.

Hint 3. $B = \bar{z}\,(\bar{A} + \frac{z}{\bar{z}}E)$, and $z/\bar{z}$ runs over the whole unit circle, while $\bar{A}$ has only finitely many eigenvalues. Choose z so that $-z/\bar{z}$ is not an eigenvalue.

Problem 8

Hint 1. By induction, derive $BA^k - b^kA^kB = a\,(1+b+\cdots+b^{k-1})A^{k-1}$ for $k \geqslant 1$. The hypotheses guarantee that $a(1 + \cdots + b^{k-1}) \neq 0$ for $1 \leqslant k \leqslant n$.

Hint 2. Suppose A is singular. Split $\mathbb{C}^n = W \oplus U$ with $W = \ker A^n \neq 0$, $U = \operatorname{Im} A^n$; both are A-invariant. Apply the identity with $k = n$ to $x \in W$: $-b^nA^nBx \in U$ equals a non-zero multiple of $A^{n-1}x \in W$, so $A^{n-1}x = 0$.

Hint 3. Descend: if $A^kW = 0$, apply the identity with exponent k to $x \in W$. Writing $Bx = w+u$, the term $A^kBx = A^ku$ lies in U, while $A^{k-1}x \in W$, so $A^{k-1}W = 0$. At the end $W = 0$, a contradiction. For B: if $b \neq 0$, rewrite the relation as $AB = -\frac{a}{b}E + \frac{1}{b}BA$, which has the same form with the roles swapped; if $b = 0$, $BA = aE$ settles it.

Problem 9

Hint 1. The minors of order $n-1$ are, up to sign, the entries of $\operatorname{adj} A$. Suppose $\det A = 0$; then $\operatorname{rank}\operatorname{adj} A \leqslant 1$.

Hint 2. If $\operatorname{adj} A = O$, all n^2 minors vanish. Otherwise $\operatorname{adj} A = uv^T$ with $u, v \neq 0$. If u has p zero coordinates and v has q, count the zero entries of uv^T: $np + nq - pq$. Since $k \geqslant 1$, $p + q \geqslant 1$, and this count is $\geqslant n > k$.

Problem 10

Hint 1. A non-zero (real) X with $AX = X(-A)$ exists if and only if A and $-A$ have a common eigenvalue (Sylvester's theorem on $AX = XB$; a complex solution gives a real one by taking real or imaginary parts). Since A is invertible, this means A has eigenvalues λ and $-\lambda$ with $\lambda \neq 0$.

Hint 2. $\operatorname{tr}\operatorname{adj} A = e_3(\lambda_1, \ldots, \lambda_4)$, the third elementary symmetric function of the eigenvalues. $A^2 + E$ is singular if and only if $\pm i$ are eigenvalues, which is such a pair.

Hint 3. Conversely, if the eigenvalues are $\lambda, -\lambda, \mu, \nu$, compute $e_1 = \mu + \nu$ and $e_3 = -\lambda^2(\mu + \nu)$. The condition $e_1 = e_3 \neq 0$ forces $\lambda^2 = -1$.

Problem 11

Hint 1. For odd n: $\det A = \det A^T = \det(-A) = (-1)^n \det A$, so $\det A = 0$.

Hint 2. For even n the determinant need not vanish (try $n = 2$). The non-zero eigenvalues of a real antisymmetric matrix are purely imaginary and come in pairs $\pm i\lambda$, so $\det A = \prod \lambda_k^2 \geqslant 0$. In fact $\det A$ is the square of a polynomial in the entries, the Pfaffian.

Problem 12

Hint 1. Write $|\det(E + A)| = \prod|1 + \lambda_k|$ over the complex eigenvalues of A, and bound each factor using $1 + x \leqslant e^x$.

Hint 2. $|1+z|^2 = 1 + 2\operatorname{Re} z + |z|^2 \leqslant e^{2\operatorname{Re} z + |z|^2}$. This gives the bound $\exp(\sum \operatorname{Re}\lambda_k + \frac{1}{2}\sum|\lambda_k|^2)$, and $\sum \operatorname{Re}\lambda_k = \operatorname{tr} A$.

Hint 3. It remains to show that $\sum|\lambda_k|^2 \leqslant \sum a_{ij}^2$. Use Schur triangularization $A = UTU^*$: the Frobenius norm is unitarily invariant and $\|T\|_F^2 \geqslant \sum|t_{kk}|^2$.

Problem 13

Hint 1. Subtract the first row from each of the others. What can you say about the entries of rows $2, \ldots, n$ now?

Hint 2. They are even, so a factor 2 comes out of each of $n-1$ rows. For sharpness, take the first row all ones and row i ($i \geqslant 2$) equal to all ones plus $2e_i$. After the subtraction its determinant is exactly 2^{n-1}.

Problem 14

Hint 1. Barbara wins. Look for a pairing strategy that forces two equal columns at the end.

Hint 2. Pair column j with column $j+1004$. Whenever Alan writes x in position (i, j), Barbara writes the same x in the paired position in the same row. Check that the paired cell is always empty.

Problem 15

Hint 1. Player 0 places only four zeros, so look for small zero patterns that force $\det = 0$ on their own: a zero row, a zero column, or a 2×2 block of zeros (rows i, j and columns k, l; then two rows are proportional).

Hint 2. Player 0 wins, and can always force one of these patterns. Up to symmetry, Player 1 opens in a corner, on an edge, or in the center. A good first reply is any cell that shares neither a row nor a column with Player 1's first 1.

Hint 3. After that, Player 0 keeps creating threats: positions where one more zero would complete a zero line or a zero 2×2 block. Player 1 has to block each threat. Go through the three cases and check that Player 0 can always make a double threat in time to complete a pattern with the fourth zero.

Problem 16

Hint 1. Changing entry (i, j) by t changes the determinant by t times the cofactor C_{ij}, since det is linear in each entry. So you need $\det A = 0$ with every cofactor non-zero, i.e. every entry of $\operatorname{adj} A$ non-zero.

Hint 2. Yes, for every n. If $\operatorname{rank} A = n-1$, then $\operatorname{adj} A = uv^T$ with $Au = 0$ and $v^T A = 0$. Pick A whose kernel and left kernel are both spanned by $(1, \ldots, 1)$, e.g. $A = nE - J$; then $\operatorname{adj} A$ is a non-zero multiple of J. ($n = 1$ is trivial.)